In a cohort spanning 91 clinical sites globally, this trial had a mere 1% Black representation equating to only three Black participants. This glaring disparity prompts questions about the effectiveness of inclusivity efforts. Historical and ongoing clinical research shows that factors such as population stratification, genetic diversity, and sociocultural differences can profoundly influence drug efficacy and safety outcomes.[3] Overlooking or marginalising a race or ethnicity can compromise the applicability and universality of the findings.

Clinical research has the esteemed role of paving the way for medical advancements. With this power, however, comes a substantial responsibility—the duty to confront and rectify systemic biases. The oversight in this study reflects broader, structural obstacles that plague clinical trials.[4] As stated before, a solution meant for all but available only to a select few is not an ideal remedy. Instead, it is a harbinger of further disparities.

The global medical community should advocate fervently for equal representation.[5] Although studies might highlight promising drugs, it is paramount to gauge their efficacy across all racial and ethnic groups adequately. True scientific triumph lies not in the breadth of discovery but in its depth of equitability.

We declare no competing interests.

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Totality of evidence refutes neoplasm risk with fezolinetant

We read with interest the Correspondence from Jonathan Douxfils and colleagues regarding an increased risk of neoplasm with the neurokinin 3 receptor antagonist fezolinetant.[1] However, important considerations warrant clarification.

The numeric imbalance identified in SKYLIGHT 4 (see appendix 4 in the full-length study paper)[2] prompted an in-depth analysis of all benign, malignant, and unspecified neoplasms from phase 2 and 3 fezolinetant studies. This analysis was combined with a comprehensive assessment of key characteristics of carcinogens,[3] an evaluation of fezolinetant structural properties and non-clinical data, and an epidemiological literature review. Given a short latency period to diagnosis, heterogeneity of tumour type, previous neoplasm or risk factor history, and presence of alternative baseline causes, a drug effect is not supported. Considering pre-existing malignancies, the US Food and Drug Administration concluded that "this brings the incidence rate of malignancy TEAEs [treatment-emergent adverse events] within the normal background rate of cancer".[4] Furthermore, no plausible mechanism exists for neurokinin 3 receptor antagonism in neoplasms. Importantly, although tachykinin agonism could be involved in neoplastic development, antagonism has been considered as a potential antineoplastic target.

Douxfils and colleagues carried out a meta-analysis with Peto odds ratios to assess the risk of neoplasms with fezolinetant.[1] Studies have shown that the Peto methodology is not appropriate for investigating rare events and is not an advocated default approach for meta-analyses.[5,6] Given different study designs and exposure durations, combining the SKYLIGHT studies into a meta-analysis is also not recommended and could lead to flawed interpretations.

In conclusion, the totality of current non-clinical and clinical data from the fezolinetant development programme does not support a signal for an increased risk of neoplasms.

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Authors' reply

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We read the Correspondence by Genevieve Neal-Perry and colleagues on our analysis reporting an increased risk of neoplasm with the neurokinin 3 receptor antagonist fezolinetant. Their conclusion is in line with the US Food and Drug Administration (FDA) Clinical Review suggesting that the incidence rate of malignancy treatment emergent adverse events is within the normal background rate of cancer. However, the FDA document states that the background incidence rate of cancer for the age group 50–59 years is 0·56%.[1] The annual rate of neoplasms with fezolinetant at 45 mg is 1·48%.[2] This prompted our in-depth analysis.

For the **sensitivity analysis** see https://osf.io/cn9re/

Neal-Perry and colleagues concluded that a drug effect is not supported on the basis of the evaluation of fezolinetant structural properties, non-clinical data, and an epidemiological literature review. They suggest that the short latency period to diagnosis, heterogeneity of tumour type, previous neoplastic or risk factor history, and presence of alternative baseline causes support their assessment.

First, any relevant pre-existing factors should be equally distributed among the different arms of the studies. A proper analysis of such pre-existing factors in the SKYLIGHT programme is not publicly available. Second, we reported a dose-dependent increase of the risk of neoplasm, suggesting that these treatment emergent adverse events could be associated with the pharmacodynamic properties of fezolinetant. Third, there is at least one potential underlying mechanism supporting a possible association with the observed increased risk of neoplasm. Indeed, neurokinin B is responsible for the release of kisspeptin. Kisspeptin can delay the metastatic cascade by preventing growth and colonisation of different metastatic cells in distant sites.[3] By antagonising the neurokinin 3 receptor, fezolinetant could alleviate the dormancy of cancer cells.[3]

Neal-Perry and colleagues mentioned that the Peto method used to compute the odds ratio is not appropriate for investigating rare events. Our use of the Peto method was guided by the Cochrane Statistical Methods Group and others who report that this method works well when intervention effects are small, events are not particularly common, and the studies have similar numbers in experimental and comparator groups.[4,5] The two other fixed-effect methods (ie, inverse variance and Mantel–Haenszel) lead to a similar conclusion (sensitivity analysis available on Open Science Framework), strengthening the robustness of our findings.

In conclusion, our data indicate an association between fezolinetant and the risk of neoplasm. The causal link of such an association is not established but we strongly believe that additional non-clinical mechanistic studies and a careful monitoring of such events with real-world evidence are needed to further characterise this risk.

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GBD 2021 Causes of Death Collaborators. Global burden of 288 causes of death and life expectancy decomposition in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. Lancet *2024; published online April 3. https://doi.org/10.1016/S0140-6736(24)00367-2*—In this Article, the key in figure 1 has been corrected, the URL for the GBD data sources has been corrected, and the appendix has been updated. These changes have been made to the online version as of April 19, 2024 and the printed version is correct.